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Review

Association of age at menopause and type 2 diabetes: A systematic review and dose-response meta-analysis of cohort studies



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ABSTRACT

Aims: Early age at menopause has been associated with increased incidence of type 2 diabetes mellitus (T2DM), but the quantitative association between age at menopause and T2DM was unclear. We performed a meta-analysis to assess the dose-response association between age at menopause and T2DM.

Methods: PubMed, Embase and Web of Science were searched up to January 5, 2019 for cohort studies that evaluated the association of age at menopause and risk of T2DM. Relative risks (RRs) and 95% confidence intervals (95% CIs) were pooled by using the random-effects models. Restricted cubic spline model was used to evaluate the linear or nonlinear relation. **Results:** We identified 6 studies for the meta-analysis (267,284 women and 19,654 cases of T2DM). The pooled RR was 0.64 (95% CI 0.44–0.94) comparing the latest with the earliest category of age at menopause. The risk of T2DM was reduced by 10% (RR=0.90, 95% CI, 0.84–0.98) with each 5-year increment in age at menopause. We found an inverse linear association between age at menopause and T2DM.

Conclusions: Our results suggest that later age at menopause was associated with lower risk of T2DM.

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Abbreviations: T2DM, type 2 diabetes mellitus; BMI, body mass index; CI, confidence interval; RR, relative risk; HR, hazard ratio; OR, odds ratio.

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1. Introduction

Type 2 diabetes mellitus (T2DM) is a major risk factor for cardiovascular diseases, with high rates of morbidity and mortality throughout the world [1]. Both the prevalence and incidence of T2DM have increased considerably in recent decades. According to the International Diabetes Federation, the number of people with diabetes is predicted to increase from 415 million in 2015 to 642 million in 2040 in the world [2]. From 2007 to 2017, the all-age disability adjusted life-years (DALYs) rates for T2DM increased by 34.2% and 33.9% for women and men [3]. Given its significant burden, identifying the risk factors is imperative for preventing T2DM.

Menopause is an important physiological period for women; it marks the end of female fertility and has a crucial influence on women's health [4]. Previous studies have shown an association of early age at menopause and increased incidence of cardiovascular disease and metabolic syndrome [5]. Studies for the effect of menopause and diabetes are insufficient, and most focused on the effect of menopausal status (pre- vs post-menopause) on T2DM but not age at menopause [4,6]. Few studies focused on the association between age at menopause and T2DM, but the findings remained inconclusive. Some have suggested no association between age at menopause and T2DM onset [7–9]. In contrast, some studies found early age at menopause associated with increased risk of T2DM [10,11], while some studies found late age at menopause was also a significant risk factor for T2DM [12,13].

In view of the inconsistent results of previous studies, we performed a dose-response meta-analysis of all available data from cohort studies of age at menopause and risk of T2DM.

2. Methods

2.1. Search strategy

We searched PubMed, Embase and Web of Science databases up to January 5, 2019 for all cohort studies investigating age at menopause and risk of T2DM. We also reviewed references lists of retrieved articles manually for additional relevant studies. The specific search strategies were in Table S1 in Supplementary material. Our analysis followed the Meta-analyses Of Observational Studies in Epidemiology (MOOSE) guidelines [14].

2.2. Study selection

Studies were eligible for inclusion if they satisfied the following criteria: (1) the study design was a cohort study; (2) the exposure of interest was age at menopause and the outcome was T2DM; (3) the investigators reported relative risks (RRs), odds ratios (ORs) or hazard ratios (HRs) with 95% confidence intervals (95% CIs); and (4) for the dose-response analysis, the authors examined at least three levels of menopausal age and provided the level-specific case and total numbers or sufficient data to calculate these data. If the same studies were repeated, studies with the largest sample size or more comprehensive information would be included in our meta-analysis.

2.3. Data extraction and quality assessment

Two reviewers (CMG and QML) extracted data and relevant information for all included studies independently by using a

standardized data collection form. The following information was extracted from each study: first author, year of publication, region, study name, study design, follow-up years, sample size, number of cases and total population in each category, RRs/ORs/HRs with 95% CIs for all categories of exposure, ascertainment method of age at menopause and T2DM, and covariates adjusted for in the multivariable analysis. If multiple estimates of the association for the outcome were reported, the estimate that adjusted for the most covariates was extracted. Any disagreement was discussed to reach consensus.

Quality was assessed by two reviewers (CMG and QML) independently. The Newcastle-Ottawa Quality Assessment Scale (NOS) [15] was used to assess the quality of cohort studies, which allowed for a possible total score of 9 (higher points indicating higher quality) summarizing 8 aspects of each study. Scores of 0–3, 4–6, and 7–9 were assigned to low, moderate, and high quality of studies, respectively.

2.4. Statistical analysis

For the present meta-analysis, RRs and 95% CIs were considered the common effect value measurement for all studies. We considered HRs or ORs from cohort studies to be directly approximate RRs [16]. We used the results of the original studies from models with the most complete adjustment of potential baseline confounders. If some data was unavailable, such as number of cases or total population in each category, we used the method provided by Xu et al. [17] to calculate approximate data based on available data. Because of the different reference categories in different studies, we unified the reference group of different studies and set the earliest age at menopause category as the reference group [18].

We used the method provided by Greenland et al. [19] to evaluate the dose-response relationship between age at menopause and T2DM. The median or mean age at menopause and the corresponding RR (95% CI) for each category were calculated for each study. If the mean or median age at menopause in each category could not be obtained, the midpoint between the upper and lower boundaries in each category was assigned. When the shortest or the longest category was open-ended, we assumed that the open-ended interval had the same width as the closest interval [20]. To explore the dose-response association between age at menopause and T2DM, study-specific RRs (95% CIs) were calculated per 5-year increment in age at menopause and then using the random-effects model to merge the study-specific RR estimates among all cohort studies. Restricted cubic spline models with 3 knots at percentiles 25%, 50% and 75% of the distribution of age at menopause were used to evaluate the potential linear or non-linear association between age at menopause and T2DM among all studies.

The heterogeneity among studies was evaluated by Cochran Q ($P \leq 0.10$ considered statistically significant) and I^2 statistic test (I^2 value 25%, 50% and 75% represented low, moderate and high heterogeneity, respectively) [21]. With $I^2 > 50\%$, subgroup analyses were further conducted to investigate potential sources of heterogeneity [22].

Sensitivity analysis was performed to evaluate the stability and reliability of this meta-analysis by excluding one study at a time. Begg's test [23] was used to test for publication bias.

All analyses involved use of STATA v11.2 (StataCorp, College Station, TX) and STATA v12.1 (StataCorp, College Station, TX). All tests were two sides and $P < 0.05$ was considered statistically significant.

3. Results

3.1. Literature search

According to the specified retrieval strategies, a total of 2010 potentially relevant articles besides 3 additional studies obtained from the reference lists of the original articles were retrieved. Finally, we obtained a total of 2013 articles. After excluding duplicates and studies that did not meet the inclusion criteria, 6 cohort studies [6,11,24–27] were available for the meta-analysis. The detailed study selection was shown in Fig. 1.

3.2. Characteristics and quality assessment of included studies

The 6 studies involved 267,284 women and 19,654 cases of T2DM in the present meta-analysis. The main characteristics of included studies are in Table 1. Four studies [11,24,25,27] were conducted in Europe. Age at menopause was self-reported by questionnaire or interview in all studies. Two studies [25,26] involved women with only natural menopause, 1 study [27] involved women with natural or surgical menopause and 3 studies [6,11,24] did not mention the type of menopause. The mean quality score was high (8.2/9) for all studies. The detailed quality score results were in Table S2 in Supplementary material.

3.3. Risk of T2DM with late versus early age at menopause

In total, 6 studies were included in the analysis of the association of age at menopause and risk of T2DM. The pooled RR for risk of T2DM in the latest age at menopause group as compared with the earliest age group was 0.64 (95% CI 0.44–0.94) with significant heterogeneity ($I^2 = 81.6\%$, $P = 0.000$; Fig. S1 in Supplementary material). We found no significant publication bias by Begg's test ($P = 0.851$).

3.4. Dose-response analysis of age at menopause and T2DM

Six studies [6,11,24–27] provided adequate data for the dose-response analysis. For each 5-year increment in age at menopause, the risk of T2DM decreased 10% (RR = 0.90, 95% CI 0.84–0.98) with high heterogeneity ($I^2 = 80.9\%$, $P = 0.000$) (Fig. 2). We found no evidence suggesting a non-linear relation between age at menopause and T2DM ($P_{\text{nonlinearity}} = 0.075$). Restricted cubic splines analysis suggested a linear negative correlation between age at menopause and T2DM (Fig. 3).

Table 1 – Characteristics of studies included in the meta-analysis.

First author (year)	Country	Average age, years	Follow-up, years	Sample size	No. of cases	Type of menopause	NOS	Covariates
Brand et al. (2013) [11]	Europe	59.2	11	7864	3691	No mention	8	Age, BMI, smoking, alcohol, physical activity, HRT education, no. of pregnancies, oral contraceptive use,
Appiah et al. (2014) [27]	US	60	9.2	2597	176	Natural or surgical menopause	9	Age, race, education, smoking, physical activity, general vascular health status (HT), reproductive factors (parity, postmenopausal estrogen use, age at menarche, age at menopause, BMI, WC
Wen et al. (2017) [6]	China	58.08	1.5	2084	104	No mention	8	Age, BMI
Muka et al. (2017) [26]	The Netherlands	66.9	9.2	3639	348	Natural menopause	9	Age, HRT, age at menarche, number of pregnancies, BMI, glucose, insulin, TC, use of lipid-lowering medication, SBP, antihypertensive medications, alcohol, smoking, education, CVD, physical activity, CRP
Pandeya et al. (2018) [25]	Multiple countries	56.1	9	126,721	4073	Natural menopause	8	Age, education, smoking, age at menarche, number of children, age at first birth, menopausal status or timing, HRT, BMI
LeBlanc et al. (2017) [24]	US	63.4	12.2	124,379	11,262	No mention	7	Age, race, BMI, HRT, physical activity, alcohol, smoking, education, marital status, number of term pregnancies, family history of diabetes, years since menopause, WC, history of oral contraceptive use and metformin use

NOS, The Newcastle-Ottawa Quality Assessment Scale; BMI, body mass index; HRT, hormone replacement therapy; HT, hypertension; WC, waist circumference; TC, total cholesterol; CVD, cardiovascular disease; SBP, systolic blood pressure; CRP, C-reactive protein.

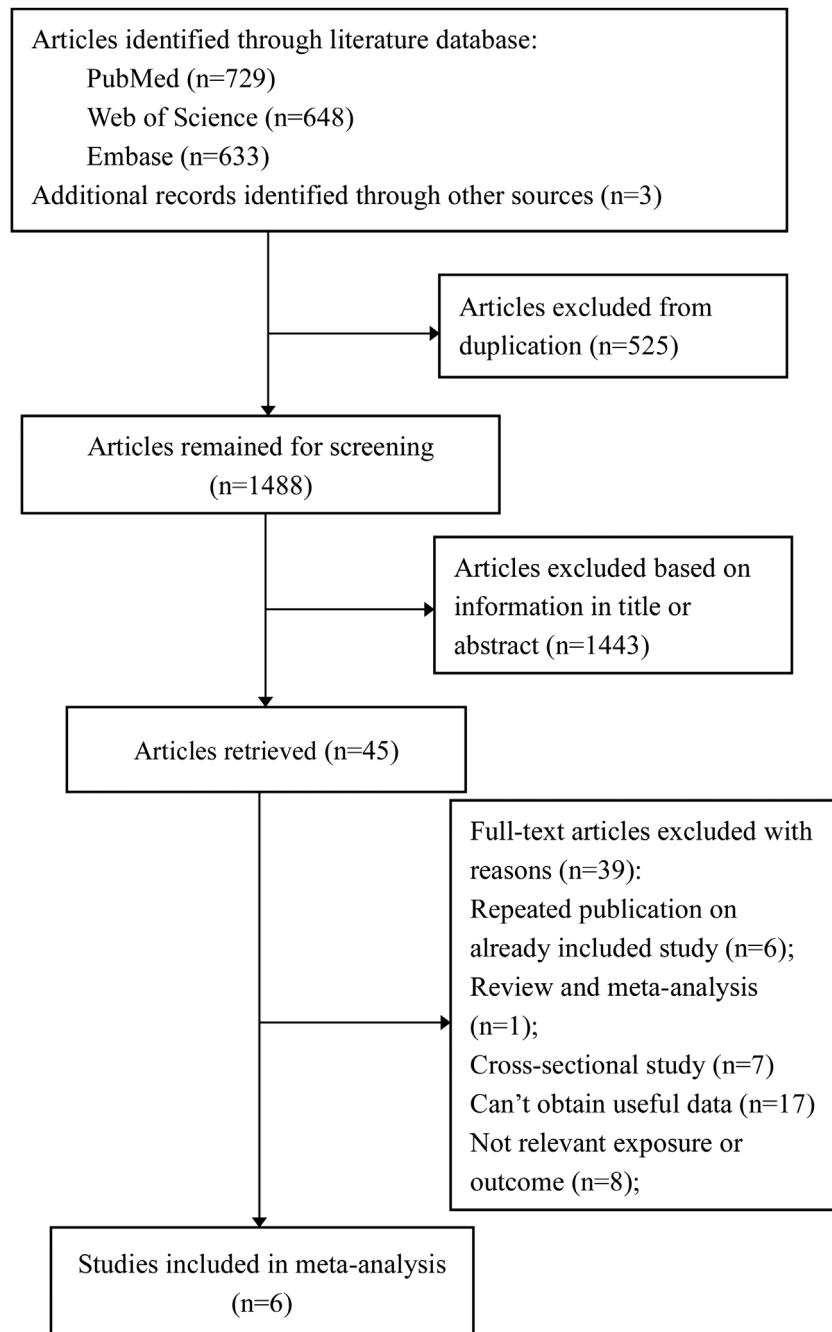


Fig. 1 – Follow chart of article selection.

3.5. Subgroup analyses

We conducted subgroup analyses to evaluate the potential sources of heterogeneity. The inverse association between age at menopause and T2DM was not observed among older women, and those with unnatural menopause and studies with follow-up more than 10 years and those studies with sample size over 5000. In general, we observed no substantial change for the association with these subgroup analyses. In stratified analyses by adjusting for age at menarche, the heterogeneity seemed lower with I^2 reduced to 58.4% and 65.5%. We did not find the sources of heterogeneity because no sig-

nificant changes of heterogeneity occurred in other subgroup analyses. Table 2 shows the subgroup analyses of the dose-response association between age at menopause and risk of T2DM.

3.6. Sensitivity analysis and publication bias

We performed sensitivity analysis to explore potential source of heterogeneity across studies and confirm the stability and robustness of our meta-analysis by sequentially omitting one study at a time. The heterogeneity decreased greatly ($I^2 = 80.9\% - 35.0\%$) when omitting one large-sample study [24],

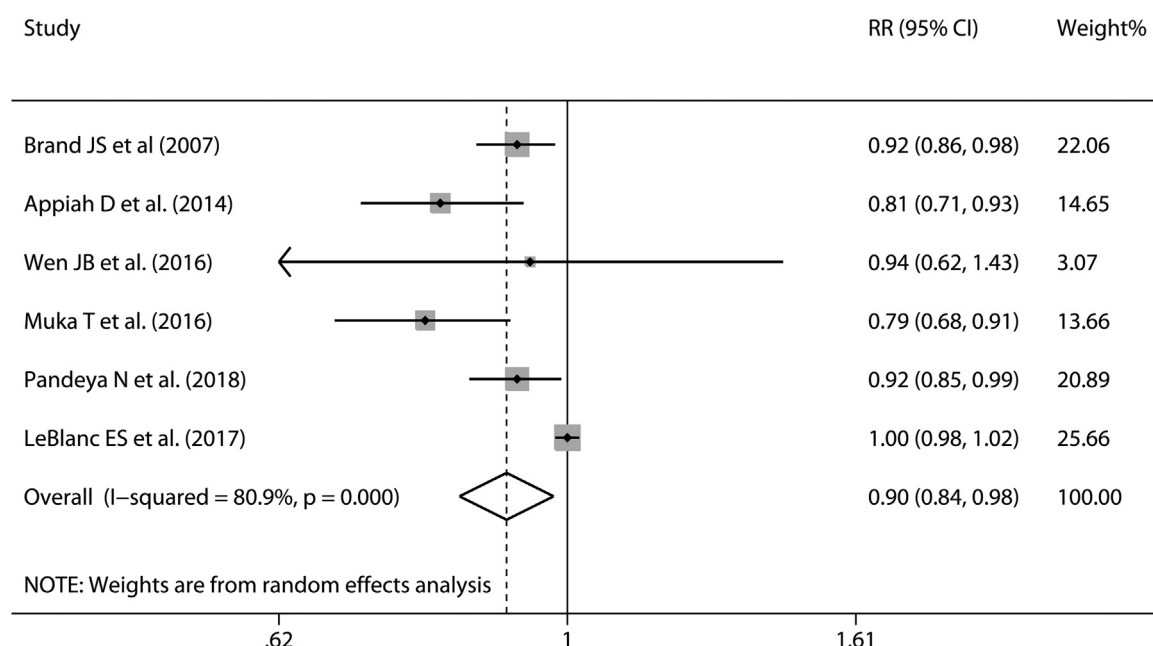


Fig. 2 – Forest plot of study-specific relative risk for type 2 diabetes mellitus per 5-year increment in age at menopause.

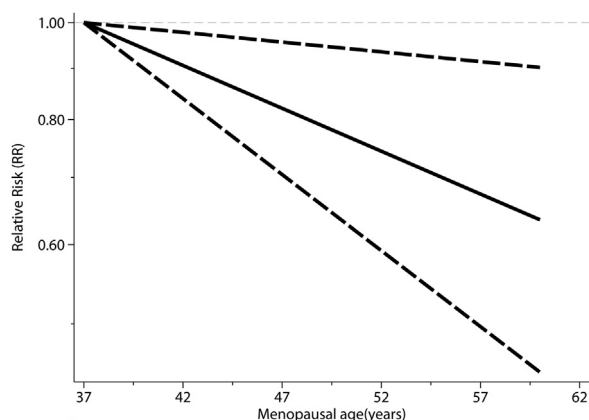


Fig. 3 – Dose-response association of age at menopause and risk of type 2 diabetes mellitus modeled by using restricted cubic splines.

but the pooled results remained consistent with the initial findings. Funnel plot analysis (not shown) and Begg's test ($P = 0.452$) revealed no significant publication bias in this meta-analysis.

4. Discussion

Results from this dose-response meta-analysis, based on 267,284 women and 19,654 cases of T2DM, indicated a significant inverse linear association between age at menopause and incidence of T2DM with a reduction of 10% in T2DM risk per 5-year increment in age at menopause. As compared with the earliest age at menopause group, the latest age at menopause group showed a 36% lower risk of T2DM (0.64, 95% CI 0.44–0.94).

There has been debate about the association between age at menopause and T2DM, and the results of epidemiological evidence remained inconsistent. To our knowledge, this is the first meta-analysis to quantify the dose-response association between age at menopause and T2DM among cohort studies. Consistent with previous observational studies [10,11], we also found that later age at menopause decrease the risk of T2DM. However, our meta-analysis differs from previous original studies because we based on cohort studies and larger sample size which increased the reliability and accuracy of the effect estimates, and further quantified the dose-response association between age at menopause and T2DM.

The mechanisms of age at menopause and T2DM remain unclear, but several underlying mechanisms may contribute to the effects. Associations between age at menopause and risk of T2DM are usually attributed to the function of the reproductive system and changes in levels of hormones, including endogenous estrogen and androgen. Bailey and Ahmed-Sorour [28] found a decline in ovarian function associated with impaired pancreatic β -cell function, which is closely related to glucose metabolism and risk of diabetes. The ovarian function of postmenopausal women further declines and eventually disappears, for significantly changed levels of sex hormones. The Heart Estrogen Progestin Replacement Study and Women's Health Initiative found that estrogen therapy could significantly lower the incidence of diabetes [29]. Similarly, a study of estrogen and diabetes [30] suggested that estrogens play an important role in preventing diabetic peripheral vascular disease. Changes in body composition after menopause are also linked to diabetes. Lipid metabolism, body composition, body mass index (BMI) and fat mass are significant risk factors for diabetes that changed rapidly in postmenopausal women [31]. The mechanism that affects the occurrence of diabetes and age at menopause is still not fully understand and needs to be further explored in future studies.

Table 2 – Subgroup dose-response analyses of the association of age at menopause per 5-year increment and risk of T2DM.

Covariates/subgroup	No. of studies	RR (95% CI)	I ² (%)	P
All studies	6	0.90 (0.84–0.98)	80.9	0.000
Region				
Europe	4	0.89 (0.81–0.99)	87.1	0.000
Other	2	0.92 (0.85–0.99)	0.0	0.921
Follow-up, years				
<10	4	0.86 (0.78–0.94)	39.1	0.178
≥10	2	0.96(0.89–1.05)	82.5	0.017
Sample size				
<5000	3	0.81 (0.73–0.89)	0.0	0.742
≥5000	3	0.95 (0.88–1.00)	78.6	0.0009
Mean age, years				
<60	3	0.92 (0.88–0.97)	0.0	0.995
≥60	3	0.87 (0.73–1.04)	89.3	0.000
Type of menopause				
Natural	2	0.89 (0.83–0.95)	69.7	0.069
Other	4	0.93 (0.84–1.02)	79.1	0.002
Adjustment				
Alcohol				
Yes	4	0.92 (0.85–0.99)	83.6	0.000
No	2	0.82 (0.72–0.93)	0.0	0.506
History of hormone replacement therapy				
Yes	4	0.89 (0.81–0.99)	87.1	0.000
No	2	0.92 (0.85–0.99)	0.0	0.921
Age at menarche			75.2	
Yes	3	0.87 (0.82–0.93)	58.4	0.090
No	3	0.97(0.90–1.04)	65.5	0.055
Family history of diabetes			82.7	
Yes	3	0.92 (0.83–1.02)	84.6	0.001
No	3	0.90 (0.85–0.95)	43.6	0.170
Physical activity				
Yes	4	0.89 (0.81–0.98)	87.1	0.070
No	2	0.92 (0.85–0.99)	0.0	0.921

RR relative risk; BMI body mass index. P for heterogeneity within each subgroup estimated by Cochran Q test.

Many factors affect age at menopause, such as cigarette smoking, education level, BMI, and form of labor. Improving the diet or changing the lifestyle can affect age at menopause to a certain extent, thereby controlling the occurrence of diabetes [32].

To discover potential sources of the high heterogeneity, we performed various subgroup analyses and sensitivity analysis. One large study significantly affected the heterogeneity [24], which suggests that it might be one of the sources of heterogeneity. The large number of participants who consisted of the observational study and the clinical trials of the Women's Health Initiative (WHI) study might account for the high heterogeneity. In addition, the identification of T2DM by self-reported medical history and medication inventory can contribute to the degree of accuracy and consistency of the results. However, we merged this study with others because the sensitivity analysis showed that the overall results did not change after removing the study, which indicates good stability and robustness of our pooled results.

Our meta-analysis has several strengths. The main strength was analyzing the dose-response relation between age at menopause and T2DM for the first time by using a

restricted cubic spline model to fit the linear association. Second, we evaluated the pooled effect of the latest versus the earliest category of age at menopause and analyzed the dose-response per 5-year increment in age at menopause. Third, subgroup analyses were performed to confirm the robustness of our results and find a possible source of heterogeneity. Finally, all studies included in the meta-analysis were cohort studies with high quality, so we could minimize the recall bias and obtained sufficient statistical power to detect the association.

This meta-analysis also has several limitations. First, few studies were included and all studies included in the meta-analysis were observational, for which the demonstration intensity of causality is low. More cohort studies are needed to further study the topic. Second, age at menopause was based on self-reports and may be subject to recall error. Finally, the women in our study were had both natural menopause and surgical or medical menopause. We did not consider the effects of surgery or medication on age at menopause and T2DM. However, the subgroup analysis by type of menopause and sensitivity analysis showed that the results of our meta-

analysis were stable and have reference value for the target population to some extent.

5. Conclusions

Our results from this meta-analysis provided quantitative evidence suggested a negative linear association between age at menopause and risk of T2DM with a restricted cubic spline model.

Declaration of interest

None.

Conflict of interest

The authors state that they have no conflict of interest.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.pcd.2019.02.001>.

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